# ETHERNET VPN
## *Standardization and Status*

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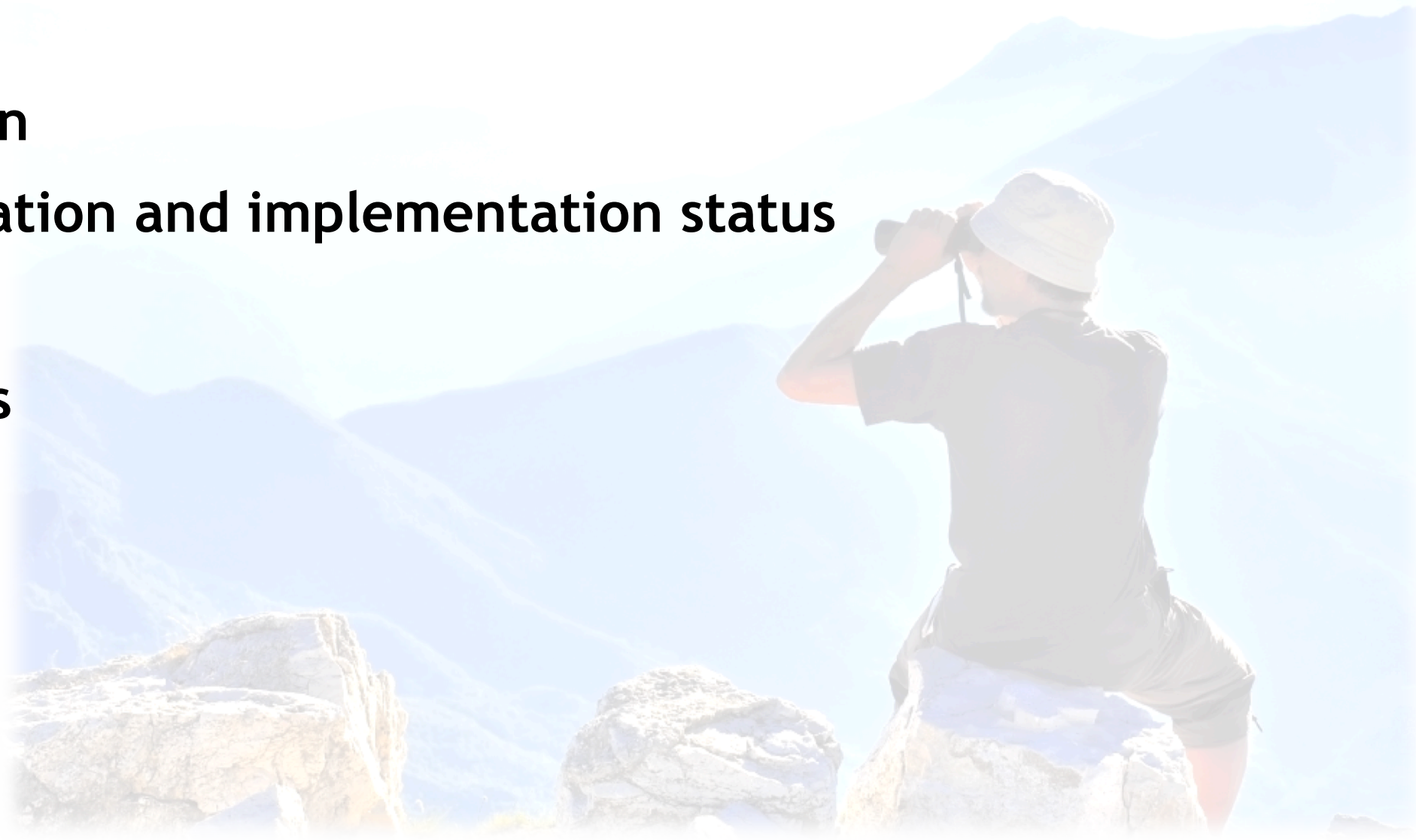
# AGENDA

1. Introduction
2. Standardization and implementation status
3. Use-cases
4. Conclusions





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# EVPN AND THE OPPORTUNITY TO MAKE IT RIGHT

- What we learned about VPNs
  - IP-VPN (RFC4364) is successfully deployed without interop issues, easy to provision, supports all-active MH but only IP traffic
  - VPLS (RFC4761/4762/6074) has control plane interop issues, provisioning vs efficiency trade-offs, flood-and-learn is not optimum, but works for any Ethernet traffic

- Why another VPN technology
  - Cloud and NFV are shifting the way networks must behave
  - EVPN is an Ethernet VPN technology (provides L2 and L3) that provides the required flexibility, it is future-proof and inherits over a decade of VPN experience

- Where can we use EVPN
  - Cloud and virtualization services
  - Integrated Layer-2 and Layer-3 VPN services
  - Overlay technologies that simplify topologies and protocols



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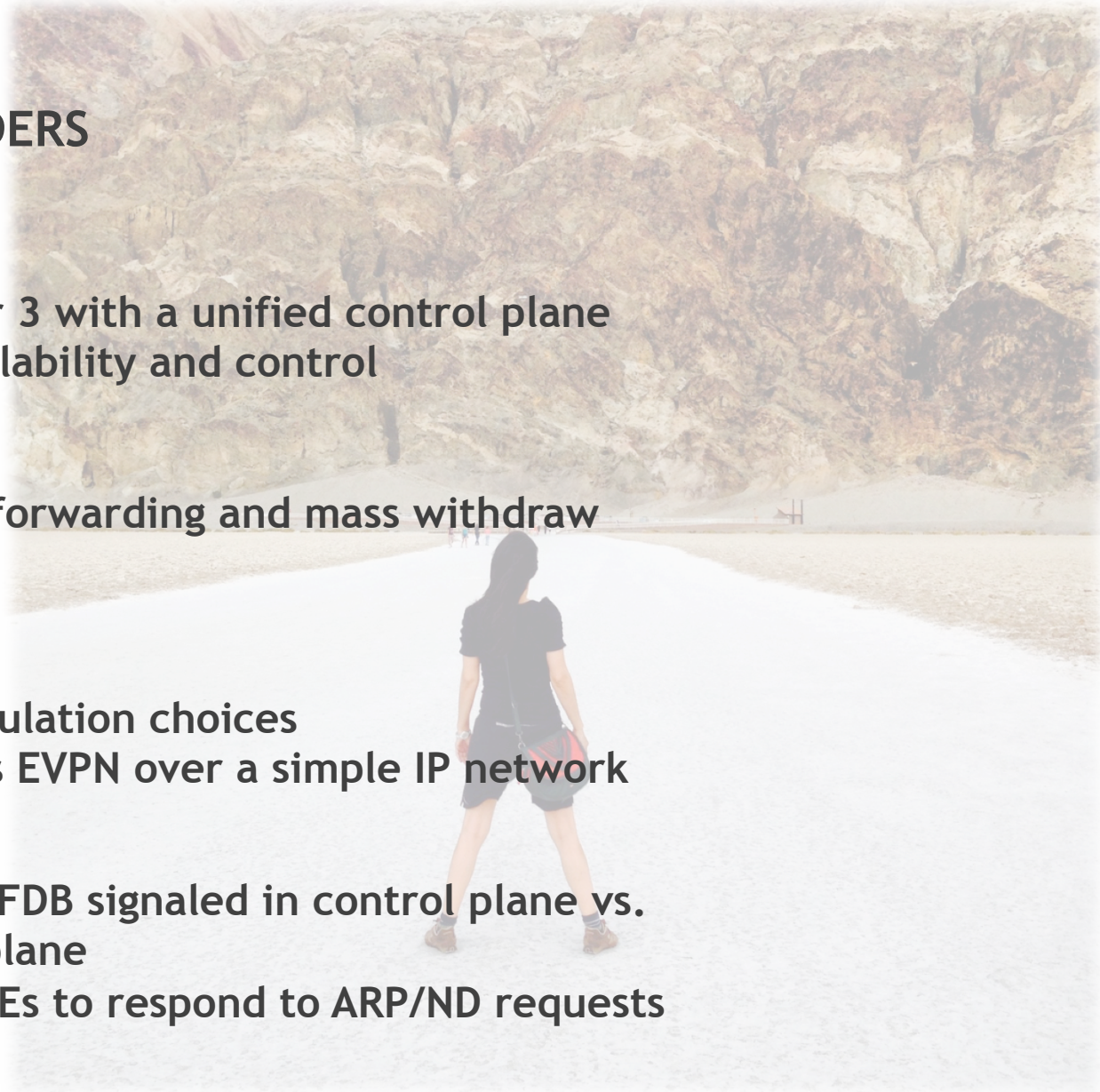
# WHAT CAN EVPN DO FOR ME?
## OPERATIONAL BENEFITS FOR SERVICE PROVIDERS



**Integrated Services**

- Delivering Layer 2 and Layer 3 with a unified control plane
- L3VPN-like operation for scalability and control

**Network Efficiency**

- Multi-homing with all-active forwarding and mass withdraw
- BUM reduction/suppression

**Design Flexibility**

- MPLS or IP data plane encapsulation choices
- VXLAN encapsulation enables EVPN over a simple IP network

**Greater Control**

- EVPN changes the paradigm: FDB signaled in control plane vs. flood-and-learn FDB in data plane
- Proxy-ARP/ND proxy allows PEs to respond to ARP/ND requests



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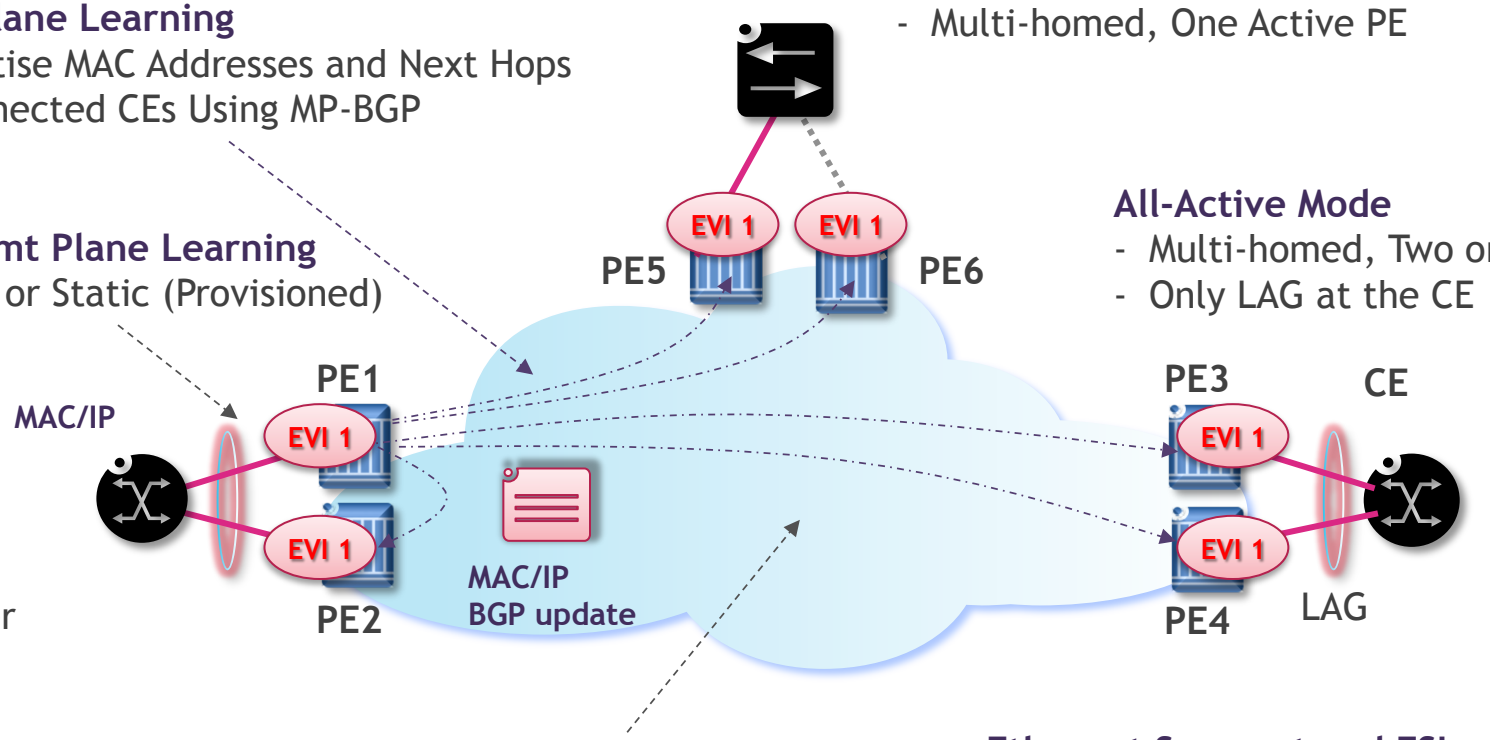
# EVPN main concepts in one shot
## *RFC7432*

**Control Plane Learning**
PEs Advertise MAC Addresses and Next Hops From Connected CEs Using MP-BGP

**Data/Mgmt Plane Learning**
Dynamic or Static (Provisioned)

**Customer Edge (CE)**
Host, VM, Router or Switch




**Single-Active Mode**
- Multi-homed, One Active PE

**All-Active Mode**
- Multi-homed, Two or More Active PEs
- Only LAG at the CE

**Data Plane Encapsulation**
RSVP-TE, LDP, BGP 3107
Segment Routing

**Ethernet Segment and ESI**
- Collection of Link(s) that Connect the CE to PEs
- ESIs are Unique Across the Network)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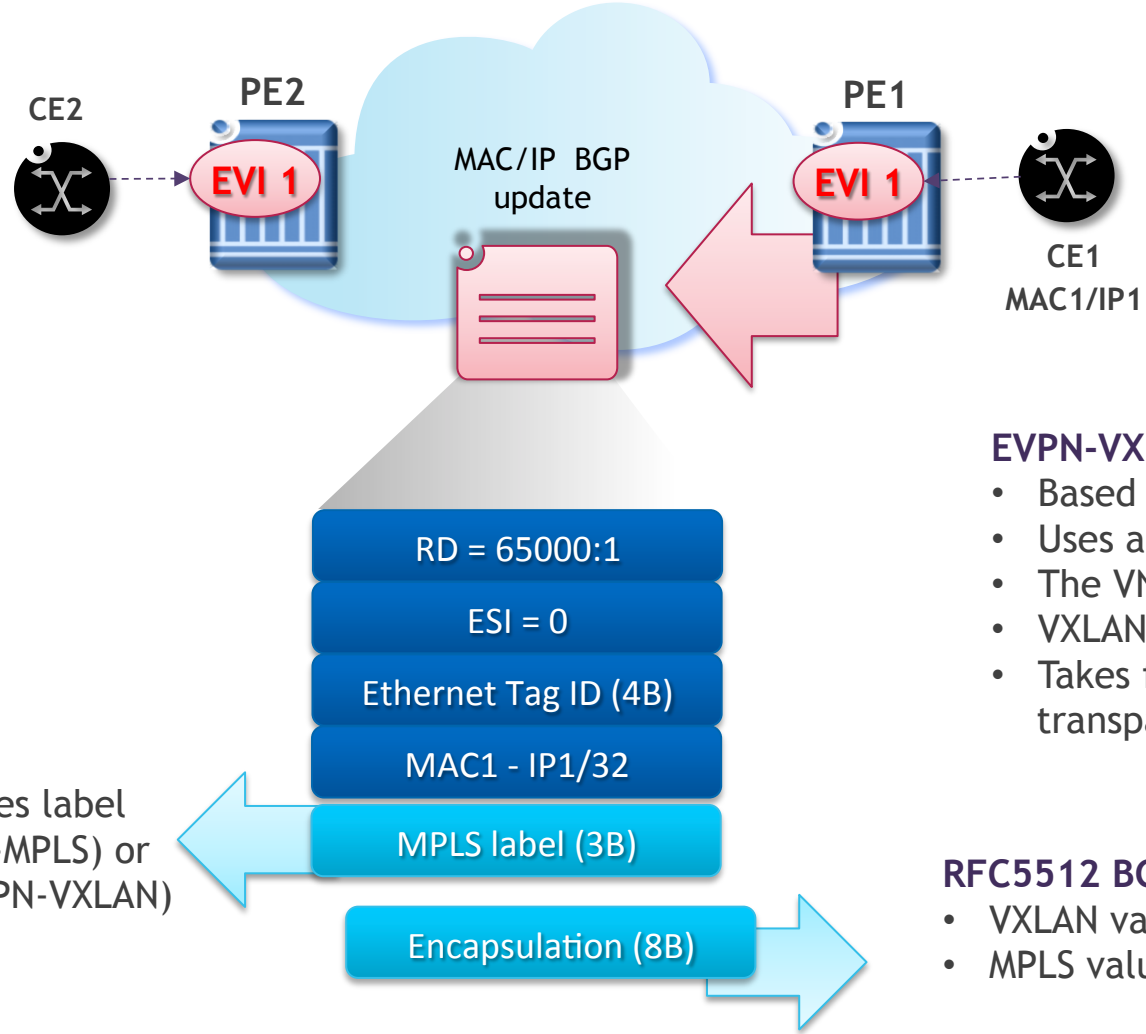
# EVPN ABSTRACTS THE CONTROL PLANE
## TO SUPPORT CURRENT AND FUTURE DATA PLANE ENCAPSULATIONS




### EVPN-MPLS
- Based on RFC7432
- Uses a 20-bit MPLS service label as MAC-VRF de-multiplexer
- Any MPLS Transport tunnel (RSVP/LDP/BGP/SR)
- Takes advantage of all the underlying MPLS capabilities (FRR, TE, etc.)

### EVPN-VXLAN
- Based on draft-ietf-bess-evpn-overlay
- Uses a 24-bit VNI as MAC-VRF de-multiplexer
- The VNI is encapsulated as part of the VXLAN header
- VXLAN is transported over UDP/IP
- Takes full advantage of the VXLAN simplicity and transparency

### RFC5512 BGP Tunnel Encapsulation
- VXLAN value = 8
- MPLS value = 10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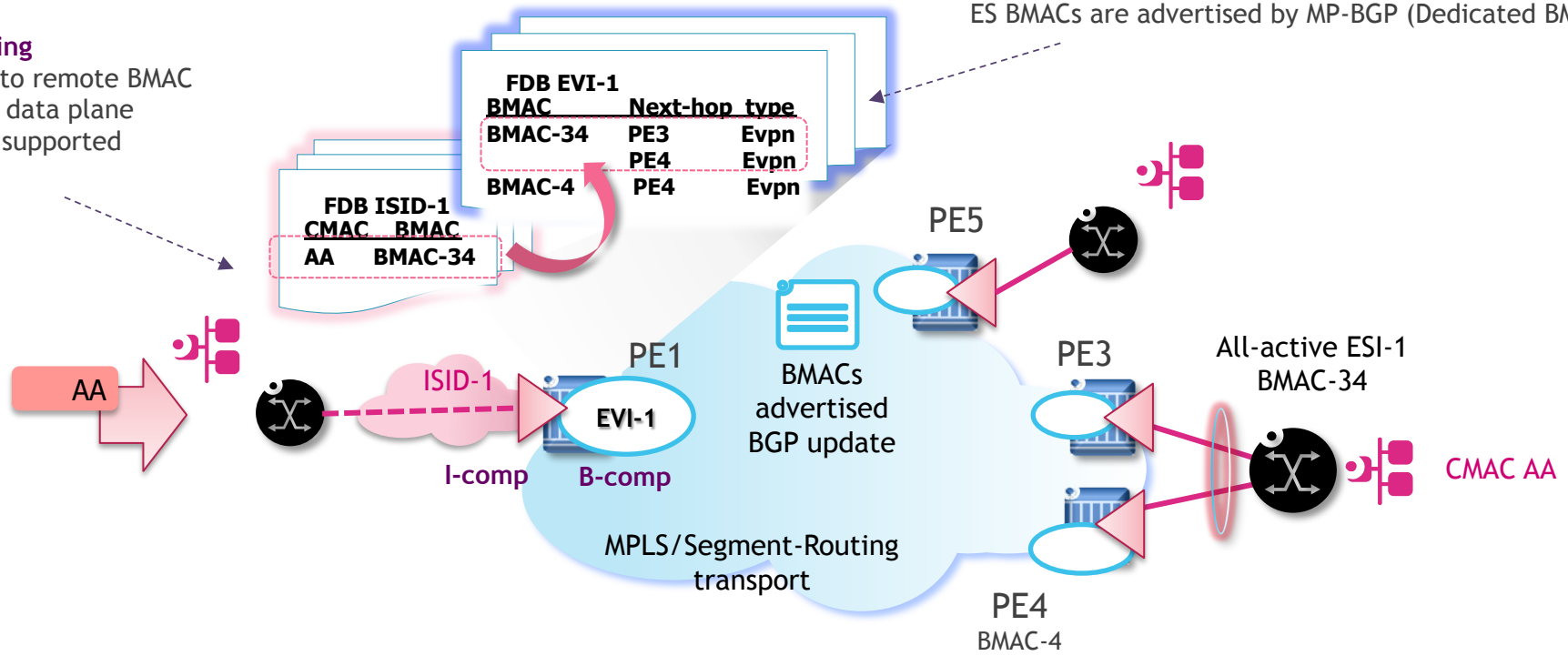
# EVPN and PBB for large L2 Service Provider networks
## PBB-EVPN (RFC7623)

**BMAC Control plane learning**
System BMACs are advertised by MP-BGP (shared BMACs)
ES BMACs are advertised by MP-BGP (Dedicated BMACs)

**CMAC Data plane learning**
CMAC-local AC or CMAC to remote BMAC mapping is learnt in the data plane
All-active multi-homing supported




**PBB-EVPN combines 802.1ah and EVPN**
- PEs have I-components mapped to B-components (EVIs)
- Reduces the number of MACs in EVPN by aggregating CMACs with BMACs

**Used to scale very large layer-2 EVPN networks**
- All EVPN Multihoming functions are supported, including single-active and all-active
- Per-ISID flooding trees are supported
- The B-component EVI uses PBB over MPLS data plane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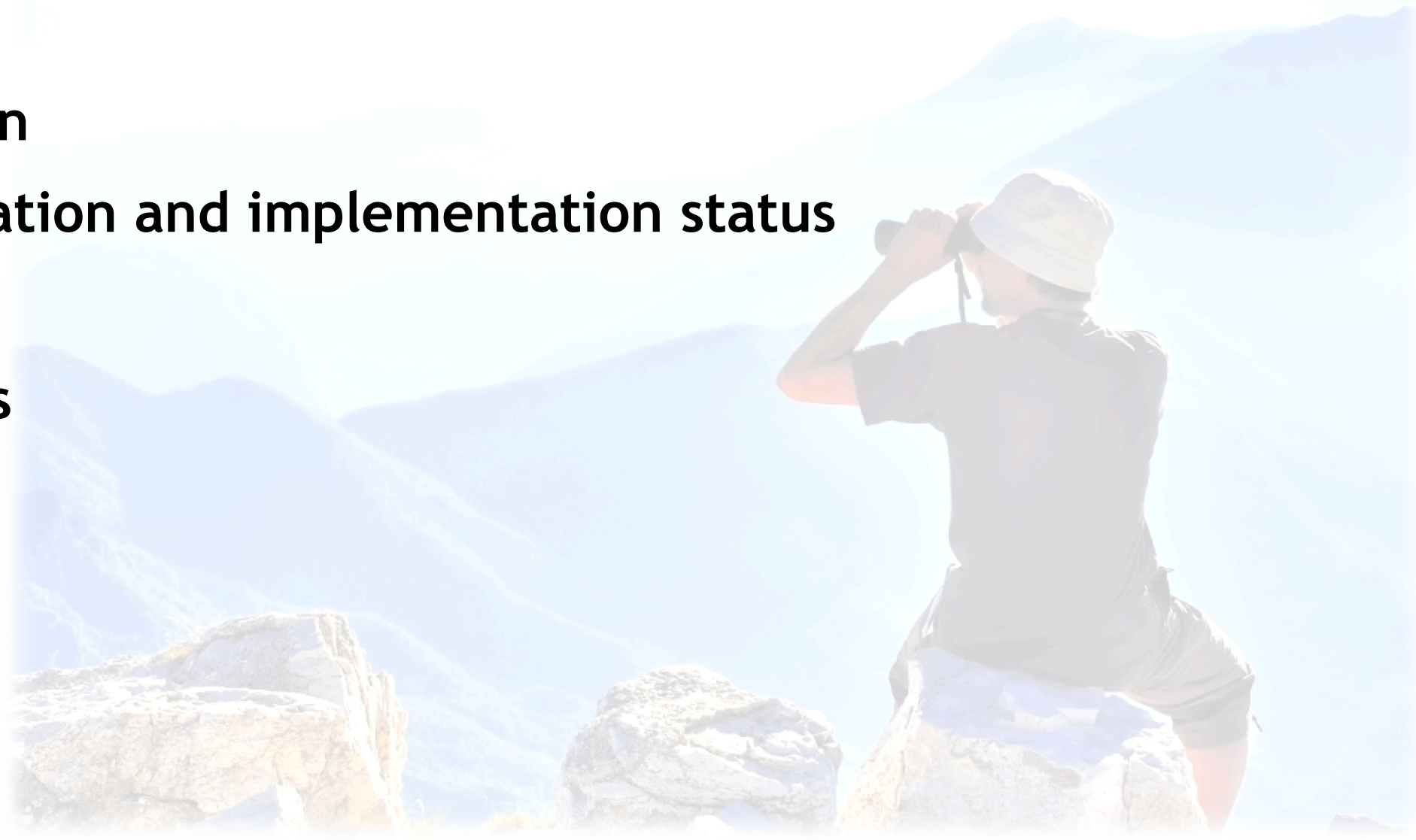



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# WHERE IS EVPN DEFINED IN THE STANDARDS?
## IETF BESS WORKING GROUP

### Layer-2 applications

draft-ietf-bess-evpn-vpws
(ELINE services)
draft-ietf-bess-evpn-etree
(ETREE services)

draft-snr-bess-proxy-arp-nd
(proxy-arp/nd for BU reduction)
draft-rabadan-bess-evpn-optimized-ir
(Optimized Ingress Replication)
draft-sajassi-bess-evpn-virt-eth-seg
(Virtual ES)

### Non-MPLS data planes

draft-ietf-bess-evpn-overlay
(EVPN for overlay tunnels)
RFC7623
(EVPN for PBB)

### Layer-3 services

draft-ietf-bess-evpn-inter-subnet-forwarding
(EVPN for inter-subnet between hosts)
draft-ietf-bess-evpn-prefix-advertisement
(ipv4/v6 prefix advertisement in EVPN)

### Integration with VPLS

draft-ietf-bess-dci-evpn-overlay
(DCI for EVPN-overlay networks)
draft-ietf-bess-evpn-vpls-seamless-integ
(Integration with VPLS and PBB-VPLS)

### DCI specific work

draft-ietf-bess-dci-evpn-overlay
(DCI Gateway for EVPN-overlay networks)
draft-ietf-bess-evpn-overlay
(DCI inter-as model B for overlay tunnels)

## EVPN - RFC 7432

- PE-CE service interfaces
- MP-BGP main routes **(ELAN services)**
- Unicast and BUM transport over MPLS tunnels
- Multi-homing Single and all-active
- Mobility Duplication Protection


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# ALU's EVPN

**2013**

EVPN-VXLAN implementation for DC

**2014**

EVPN-VXLAN integrated layer-2/layer-3

**2015**

Feb'15 - First EVPN-VXLAN interop demo at EANTC (Cisco/Juniper/ALU/Nuage)

EVPN-MPLS and PBB-EVPN Including single-active and all-active MH

**2016**

EVPN-VPWS, EVPN mLDP, EVPN AR, EVPN-ETREE, EVPN-MPLS/VXLAN integration and much more!



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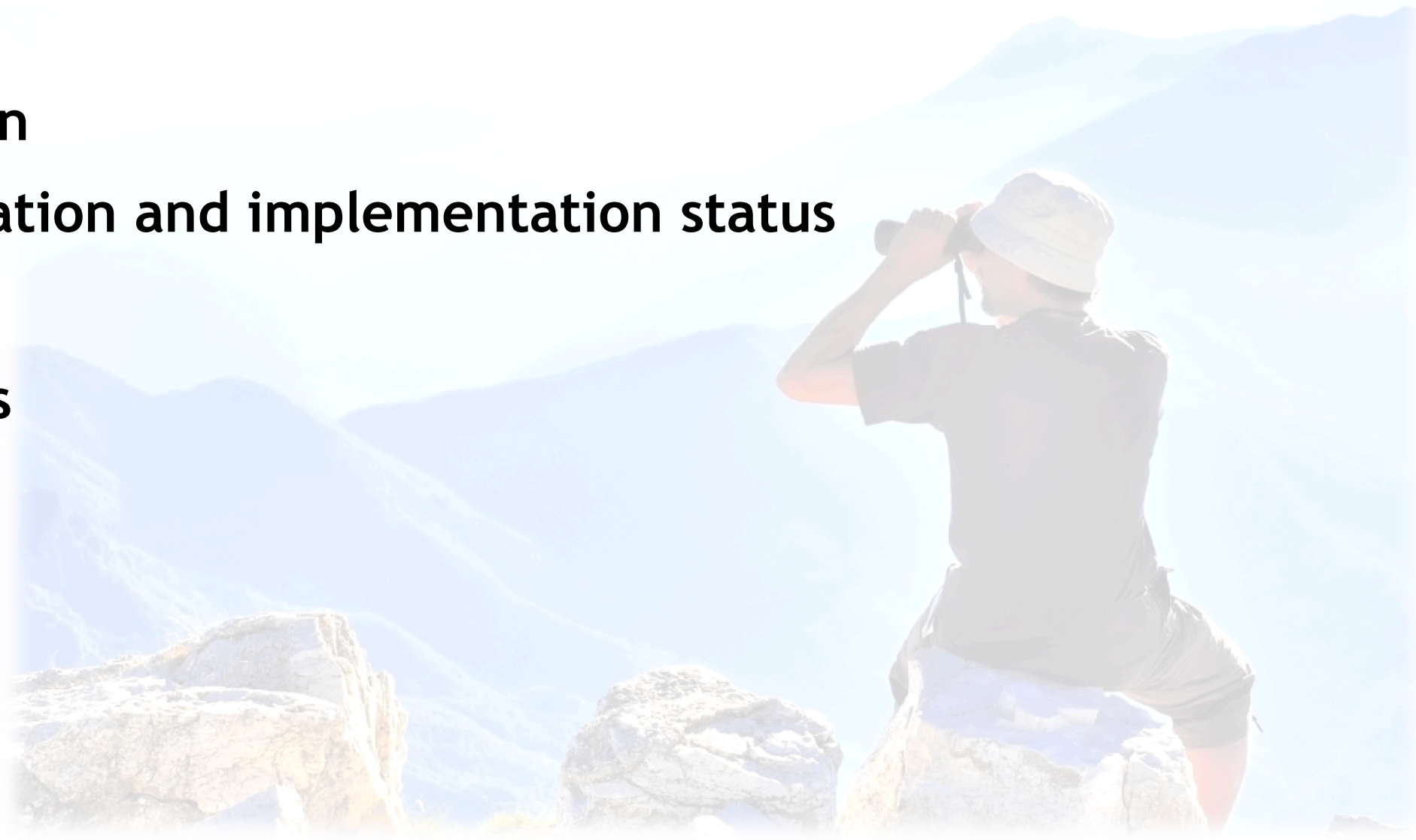



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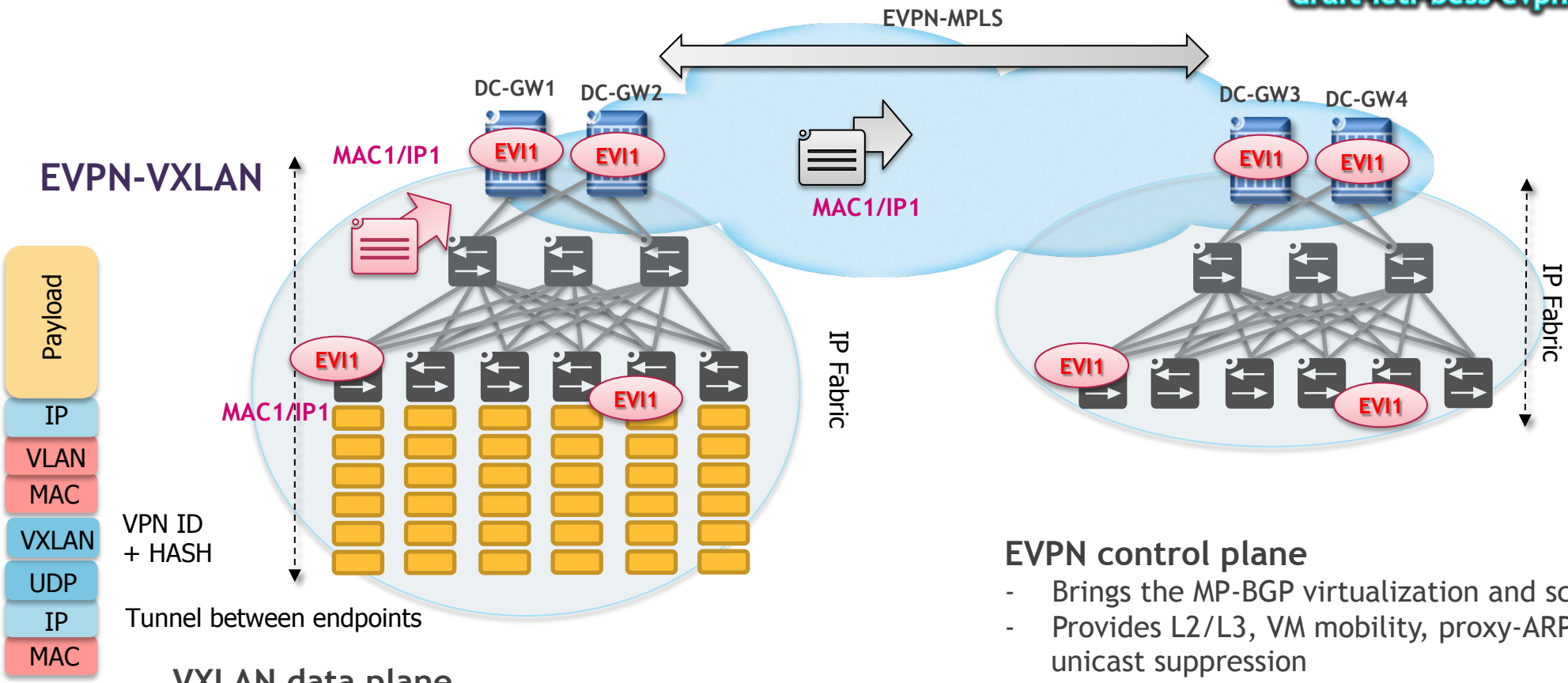
# Data Center and DCI use-case

draft-ietf-bess-evpn-overlay
draft-ietf-bess-dci-evpn-overlay
draft-rabadan-bess-evpn-optimized-ir
draft-ietf-bess-evpn-inter-subnet-forwarding
draft-ietf-bess-evpn-prefix-advertisement




## EVPN control plane

- Brings the MP-BGP virtualization and scalability to the DC
- Provides L2/L3, VM mobility, proxy-ARP/ND, ARP/unknown unicast suppression

VXLAN

## VXLAN data plane

- Provides the required scalability, performance and simplicity
- De-facto standard with assisted hardware in servers
- ECMP and fast resiliency



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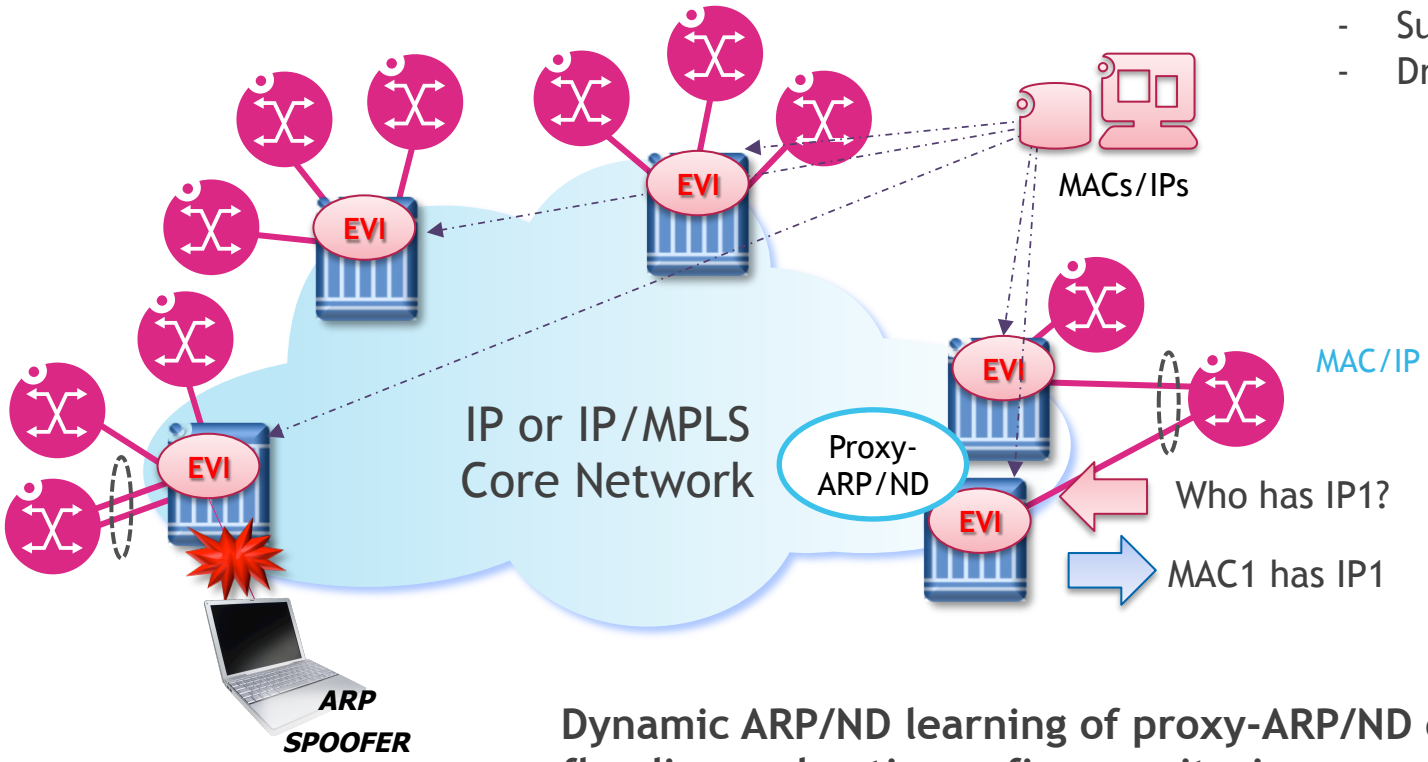
# INTERNET EXCHANGE POINTS (IXP)
## PEERING FABRIC

draft-snr-bess-evpn-proxy-arp-nd




**Static MAC/IP provisioning of the router interfaces for maximum security**

- Suppresses unknown and ARP/ND flooding
- Drops unknown source MACs

**EVPN required features**

- L2 interconnection over a VXLAN or MPLS peering fabric
- Proxy-ARP/ND and unknown/ARP/ND suppression
- MAC duplication, MAC protection
- Anti-spoofing operation

**Dynamic ARP/ND learning of proxy-ARP/ND entries for easy provisioning, minimum flooding and anti-spoofing monitoring**

- Dynamic learning of ARP/ND entries is possible
- Anti-spoofing monitors hosts claiming the same IP
  - If a duplicate is detected, an alarm is triggered and MAC/IPs put in hold-down mode
  - An option to inject an anti-spoof mac is possible too



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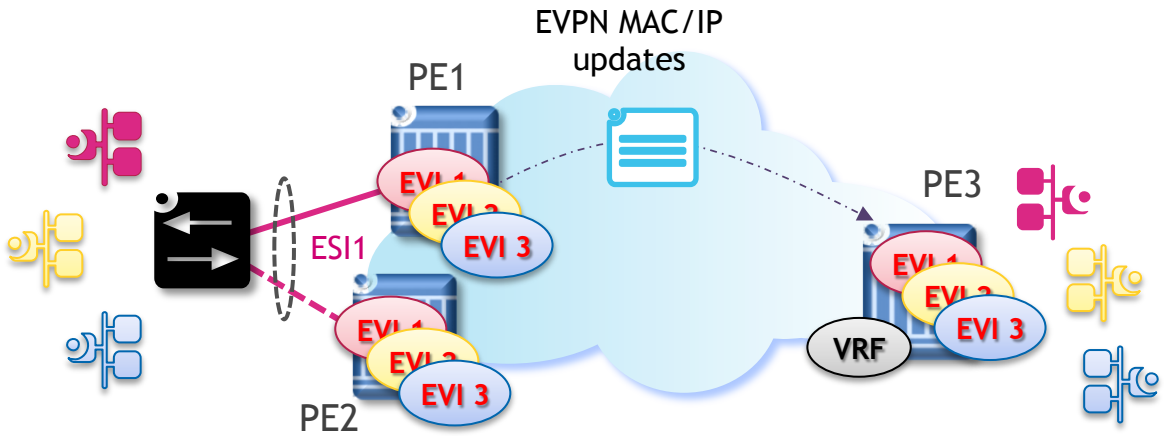
# PROVIDER-PROVISIONED VPNS
## LAYER-2 AND LAYER-3 SERVICES

RFC7432, 7623
draft-ietf-bess-evpn-etree
draft-ietf-bess-evpn-vpws
draft-ietf-bess-evpn-inter-subnet-forwarding
draft-ietf-bess-evpn-prefix-advertisement
draft-sajassi-bess-evpn-virt-eth-seg
draft-ietf-bess-evpn-vpls-seamless-integ




### Required EVPN features
- IP-prefix advertisement and inter-subnet forwarding
- Multi-homing:
  - All-active multihoming for link utilization
  - Single-active multihoming for better determinism
  - Virtual ES for access Ethernet and MPLS networks
- PBB-EVPN for large layer-2 VPNs

### EVPN provides layer-2 and layer-3 services
- Both services are provided through the same logical AC to the customer
- Layer-2 services include: ELAN, ELINE (EVPN-VPWS) and ETREE (EVPN-ETREE)
- One VPN technology for all services, no need for multiple protocols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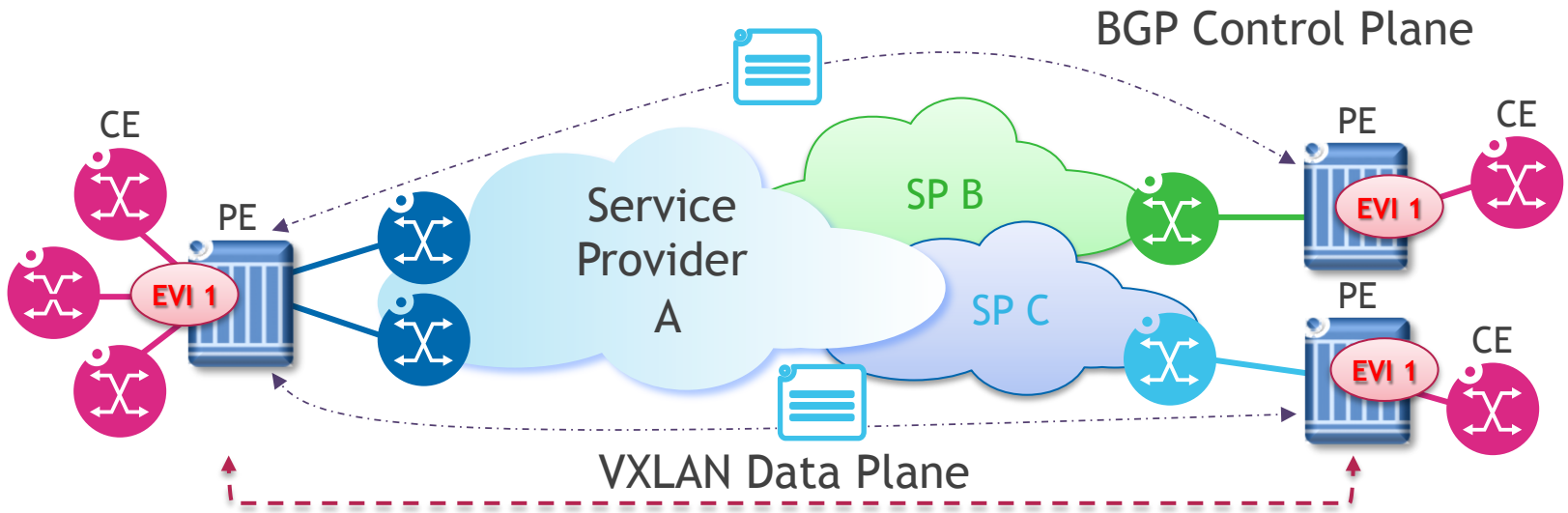
# ENTERPRISE-PROVISIONED OVERLAY VPNS




**EVPN-VXLAN works over any IP service to provide a flexible Layer-2 and Layer-3 VPN**

- Just requires IP connectivity between the sites, no need to run any MPLS or special configuration by the IP service provider
- Service Provider is transparent to EVPN
- EVPN overlay is transparent to service providers

**VPN routing between endpoints can be controlled with BGP (ipv4) and routing policies to service providers**

**Routing and MAC/IP advertisements within EVPN controlled via iBGP (evpn) between PEs**



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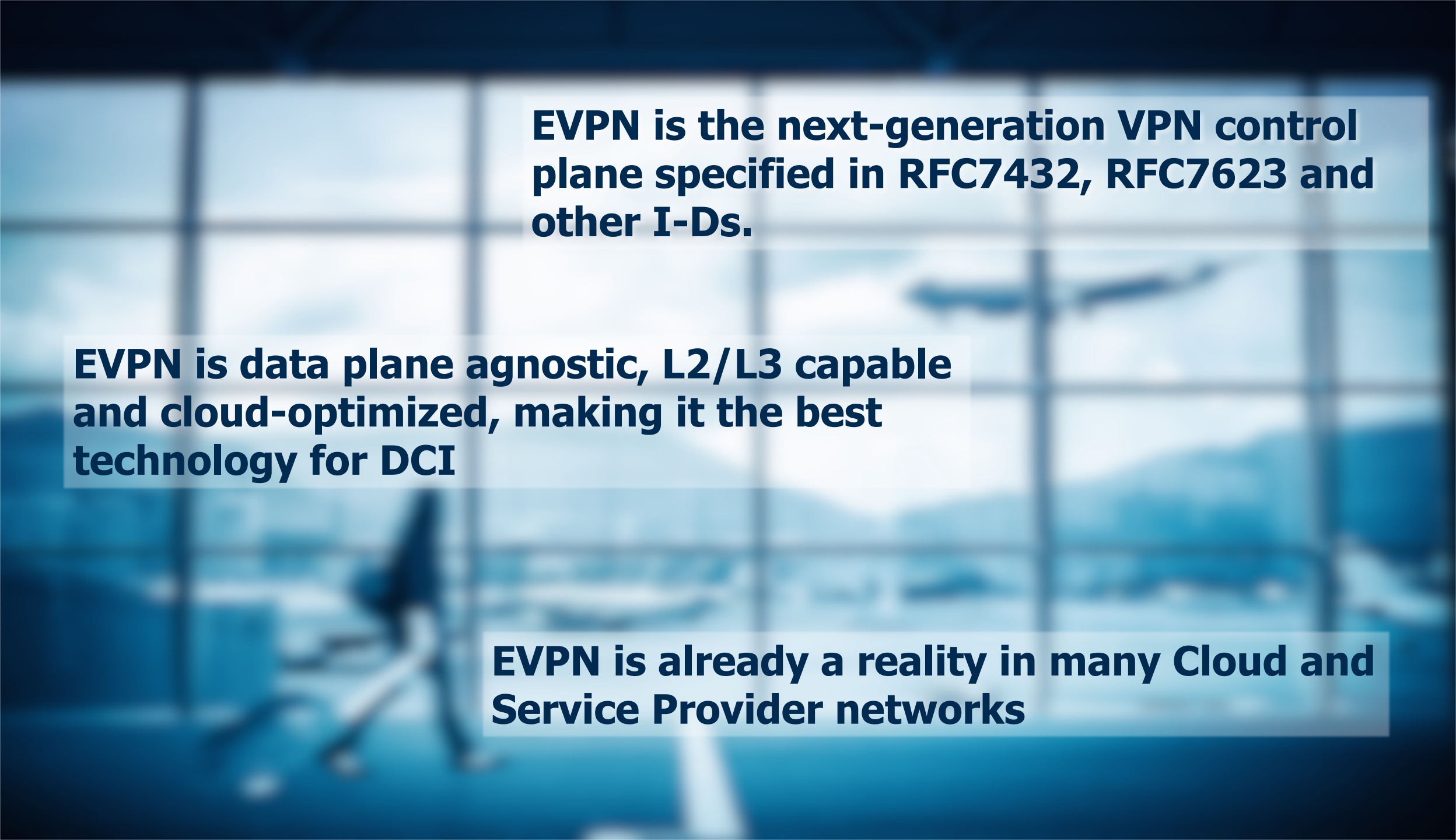
EVPN is the next-generation VPN control plane specified in RFC7432, RFC7623 and other I-Ds.
EVPN is data plane agnostic, L2/L3 capable and cloud-optimized, making it the best technology for DCI
EVPN is already a reality in many Cloud and Service Provider networks